Cloud-containerization-and-DevOps

Experiment 10: Orchestration via Docker Swarm

Introduction

The Progression Path

```
docker run  →  Docker Compose  →  Docker Swarm  →  Kubernetes
   |               |                  |                |
Single container  Multi-container    Orchestration    Advanced
                 (single host)       (basic)         orchestration
```

> **This experiment focuses on:** Moving from Compose → Swarm

Hands-On

Step-1:- Create a `docker-compose.yml`

```
nano docker-compose.yml
```

Paste This:

```yaml
version: '3.9'

services:
  db:
    image: mysql:5.7
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql
```

```yaml
  wordpress:
    image: wordpress:latest
    depends_on:
      - db
    ports:
      - "8080:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html

volumes:
  db_data:
  wp_data:
```

palakagarwal_16@LAPTOP-2S

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ cat docker-
compose.yml
version: '3.9'

services:
  db:
    image: mysql:5.7
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    depends_on:
      - db
    ports:
      - "8080:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html

volumes:
  db_data:
  wp_data:
```

Step-2:- Initialize Swarm

```
docker swarm init
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker swar
m init
Swarm initialized: current node (vun5tkvap6dlkk6blb8c3l21n) is now a manager
.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-2wf4ugl3r42j2knlulxvc0igz8apjvd5azq9l
3l0rpg174q5fm-3bgwurkbv87gln9fmhc5xyfx0 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and fo
llow the instructions.
```

Step-3:- Verify Swarm is Active

```
docker node ls
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker node
 ls
ID                            HOSTNAME         STATUS    AVAILABILITY   MANA
GER STATUS   ENGINE VERSION
vun5tkvap6dlkk6blb8c3l21n *   docker-desktop   Ready     Active         Lead
er           29.2.1
```

Step-4:- Deploy as a stack

```
docker stack deploy -c docker-compose.yml wpstack
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker stac
k deploy -c docker-compose.yml wpstack
Ignoring unsupported options: restart

Since --detach=false was not specified, tasks will be created in the backgro
und.
In a future release, --detach=false will become the default.
Creating network wpstack_default
Creating service wpstack_db
Creating service wpstack_wordpress
```

Step-5:- Verify by listing Services

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker serv
ice ls
ID             NAME                MODE         REPLICAS   IMAGE
  PORTS
ybka4kvtu2e5   wpstack_db          replicated   1/1        mysql:5.7

on3fj4rpmkxz   wpstack_wordpress   replicated   0/1        wordpress:latest
  *:8080->80/tcp
```

Step-6:- Details for container service

```
docker service ps wpstack_wordpress
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker serv
ice ps wpstack_wordpress
ID             NAME                   IMAGE              NODE            DES
IRED STATE   CURRENT STATE              ERROR     PORTS
qn0kyckh0au4   wpstack_wordpress.1   wordpress:latest   docker-desktop   Run
ning         Preparing 20 seconds ago
```

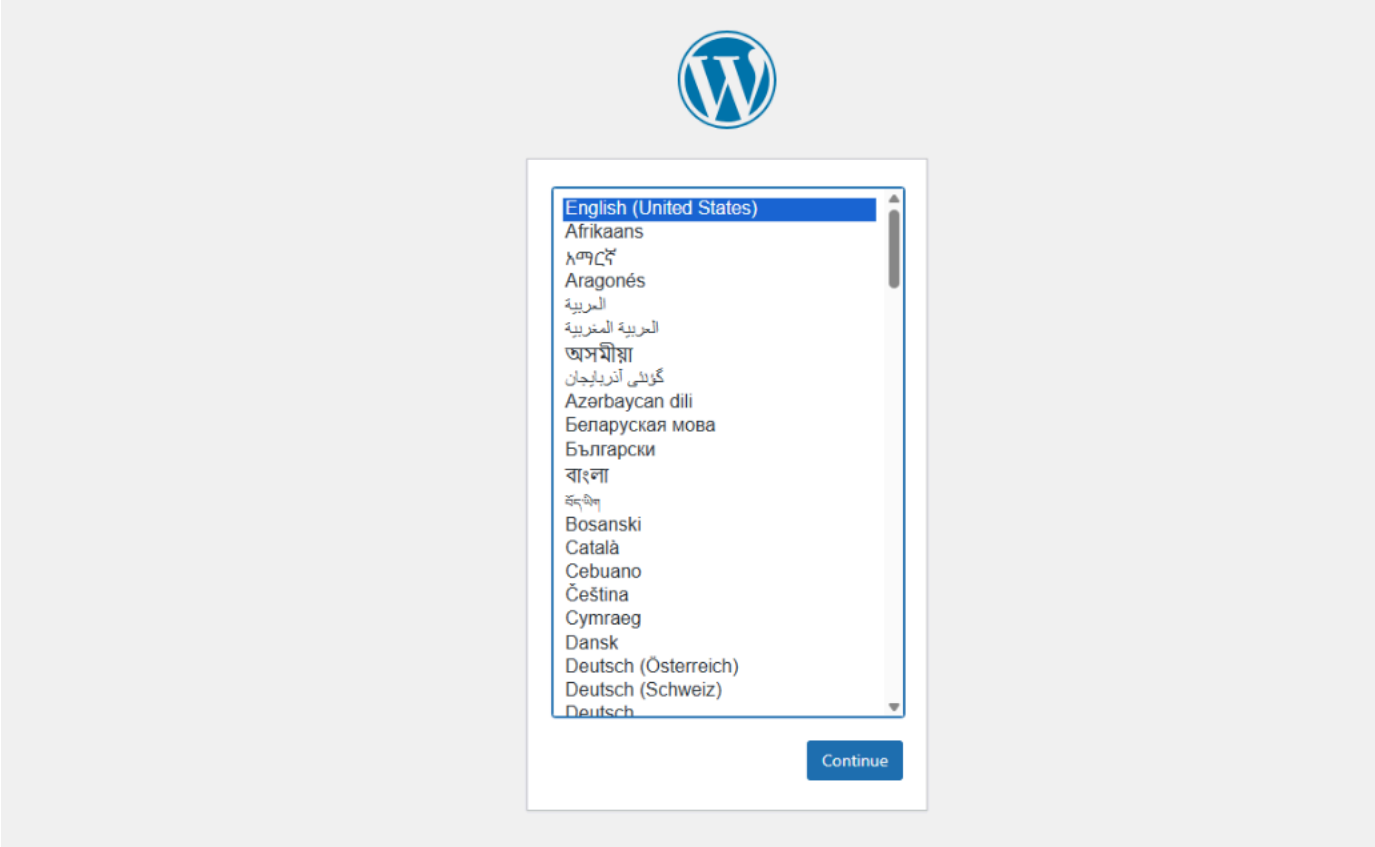
Step-7:- List Containers

```
docker ps
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
          STATUS                          PORTS
                                                                   NAMES
6c1ffef4a395   mysql:5.7                      "docker-entrypoint.s…"   28 seco
nds ago   Up 28 seconds                   3306/tcp, 33060/tcp
                                                                   wpstack_db.1.z7898s
85girsyfmj1mm0i11l7
3d5a2f94a432   sonarqube:lts-community        "/opt/sonarqube/dock…"   5 days
ago       Up 3 minutes                    0.0.0.0:9001->9000/tcp, [::]:9001-
>9000/tcp                                                          sonarqube
de3235249e19   postgres:13                    "docker-entrypoint.s…"   5 days
ago       Up 3 minutes                    5432/tcp
                                                                   sonar-db
16caf86b6b3a   portainer/portainer-ce:lts     "/portainer"             2 weeks
 ago      Up 3 minutes                    0.0.0.0:8000->8000/tcp, [::]:8000-
>8000/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp   portainer
2d80294c3c52   docker-lab-app                 "docker-entrypoint.s…"   2 weeks
 ago      Restarting (1) 40 seconds ago
                                                                   multi-stage-app
84697fc912c5   postgres:15                    "docker-entrypoint.s…"   2 weeks
 ago      Up 3 minutes                    5432/tcp
```

Step-8:- Verify application on Containers

```
http://localhost:8080/
```



Step-9:- Scale application Containers

```
docker service scale wpstack_wordpress=3
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker serv
ice scale wpstack_wordpress=3
wpstack_wordpress scaled to 3
overall progress: 3 out of 3 tasks
1/3: running
2/3: running
3/3: running
verify: Service wpstack_wordpress converged
```

Step-10:- Verify Scalling

```
docker service ls
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker serv
ice ls
ID             NAME                MODE         REPLICAS   IMAGE
  PORTS
ybka4kvtu2e5   wpstack_db          replicated   1/1        mysql:5.7

on3fj4rpmkxz   wpstack_wordpress   replicated   3/3        wordpress:latest
  *:8080->80/tcp
```

Step-11:- List Wordpress Containers

```
docker ps | grep wordpress
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker ps |
 grep wordpress
981cec7eeda0   wordpress:latest             "docker-entrypoint.s…"   About a
 minute ago   Up About a minute              80/tcp
                                                                wpstack_wordpre
ss.3.tf0cz1ja5uxytaww799ev3vya
72238897389d   wordpress:latest             "docker-entrypoint.s…"   About a
 minute ago   Up About a minute              80/tcp
                                                                wpstack_wordpre
ss.2.vdq51w30fiqgv07js0zia0wyo
bb9d8ab0e61d   wordpress:latest             "docker-entrypoint.s…"   About a
 minute ago   Up About a minute              80/tcp
                                                                wpstack_wordpre
ss.1.qn0kyckh0au4mxd1ebroi647z
```

Step-12:- Kill any Container

```
docker kill <container-id>
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker kill
 981cec7eeda0
981cec7eeda0
```

Step-13:- Watch Swarm Recreating Contrainer

```
docker service ps wpstack_wordpress
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker serv
ice ps wpstack_wordpress
ID             NAME                     IMAGE              NODE
 DESIRED STATE   CURRENT STATE               ERROR
PORTS
qn0kyckh0au4   wpstack_wordpress.1       wordpress:latest   docker-desktop
 Running         Running about a minute ago
vdq51w30fiqg   wpstack_wordpress.2       wordpress:latest   docker-desktop
 Running         Running about a minute ago
rgk0u908g65n   wpstack_wordpress.3       wordpress:latest   docker-desktop
 Running         Running 12 seconds ago
tf0cz1ja5uxy    \_ wpstack_wordpress.3   wordpress:latest   docker-desktop
 Shutdown        Failed 18 seconds ago        "task: non-zero exit (137)"
```

Step-14:- Verify Container running

```
docker ps | grep wordpress
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker ps |
 grep wordpress
15eb0540dfaf   wordpress:latest            "docker-entrypoint.s…"   23 seco
nds ago   Up 17 seconds                   80/tcp
                                                                 wpstack_wordpress.3
.rgk0u908g65nsk21fmuxlr5r5
72238897389d   wordpress:latest            "docker-entrypoint.s…"   2 minut
es ago    Up 2 minutes                    80/tcp
                                                                 wpstack_wordpress.2
.vdq51w30fiqgv07js0zia0wyo
bb9d8ab0e61d   wordpress:latest            "docker-entrypoint.s…"   2 minut
es ago    Up 2 minutes                    80/tcp
                                                                 wpstack_wordpress.1
.qn0kyckh0au4mxd1ebroi647z
```

Step-15:- Remove Stack

```
docker stack rm wpstack
```

```
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker stac
k rm wpstack
Removing service wpstack_db
Removing service wpstack_wordpress
Removing network wpstack_default
```

Step-16:- Verify Service Removal

```
docker service ls
docker ps
```

```
docker ps
ID        NAME      MODE      REPLICAS   IMAGE     PORTS
CONTAINER ID   IMAGE                        COMMAND                  CREATED
          STATUS                         PORTS
                                                        NAMES
15eb0540dfaf   wordpress:latest             "docker-entrypoint.s…"   32 seco
nds ago   Up 27 seconds                   80/tcp
                                                        wpstack_wordpress.3
.rgk0u908g65nsk21fmuxlr5r5
72238897389d   wordpress:latest             "docker-entrypoint.s…"   2 minut
es ago    Up 2 minutes                    80/tcp
                                                        wpstack_wordpress.2
.vdq51w30fiqgv07js0zia0wyo
bb9d8ab0e61d   wordpress:latest             "docker-entrypoint.s…"   2 minut
es ago    Up 2 minutes                    80/tcp
                                                        wpstack_wordpress.1
.qn0kyckh0au4mxd1ebroi647z
6c1ffef4a395   mysql:5.7                    "docker-entrypoint.s…"   2 minut
es ago    Up 2 minutes                    3306/tcp, 33060/tcp
                                                        wpstack_db.1.z7898s
85girsyfmj1mm0i11l7
3d5a2f94a432   sonarqube:lts-community      "/opt/sonarqube/dock…"   5 days
ago       Up 6 minutes                    0.0.0.0:9001->9000/tcp, [::]:9001-
>9000/tcp                                               sonarqube
de3235249e19   postgres:13                  "docker-entrypoint.s…"   5 days
ago       Up 6 minutes                    5432/tcp
                                                        sonar-db
16caf86b6b3a   portainer/portainer-ce:lts   "/portainer"             2 weeks
 ago      Up 6 minutes                    0.0.0.0:8000->8000/tcp, [::]:8000-
>8000/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp   portainer
2d80294c3c52   docker-lab-app               "docker-entrypoint.s…"   2 weeks
 ago      Restarting (1) 55 seconds ago
                                                        multi-stage-app
84697fc912c5   postgres:15                  "docker-entrypoint.s…"   2 weeks
 ago      Up 6 minutes                    5432/tcp
                                                        postgres-db
palakagarwal_16@LAPTOP-2SDBVCPO:/mnt/c/Users/hp/wordpress-swarm$ docker serv
ice ls
ID        NAME      MODE      REPLICAS   IMAGE     PORTS
```

Conclusion

Summary

You started with	You can now do
Single container (`docker run`)	Multi-container (Compose)
Manual scaling	One-command scaling (`scale`)
Manual recovery	Automatic self-healing
Single host	Multi-host cluster ready

Final Takeaway

> **Compose defines the application. Swarm runs it reliably.**

Quick Reference Card

```
# Initialize Swarm
docker swarm init

# Deploy stack
docker stack deploy -c docker-compose.yml <stack-name>

# List services
docker service ls

# Scale service
docker service scale <stack-name_service-name>=<replicas>

# See service tasks
docker service ps <service-name>

# Remove stack
docker stack rm <stack-name>

# Leave Swarm (if needed)
docker swarm leave --force
```